

Predicting ages of acquisition for children’s early vocabulary across 27 languages and dialects

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Abstract

What factors contribute to the growth of children’s early vocabulary? One method for exploring this question is investigating predictors (e.g., frequency) that differentiate words learnt earlier from those learnt later. A more comprehensive account requires the examination of multiple language families and multiple levels of linguistic representation (e.g., phonological, morphosyntactic, semantic). Here, we studied 10 predictors of word ages of acquisition across 27 languages and dialects. We found that words that were more frequent, concrete, and associated with babies were learnt earlier, whereas words that had greater length in phonemes and mean length of utterance were learnt later. There was no reliable effect of other morphosyntactic predictors, or of phonological neighbourhood. We found evidence of a tradeoff between a language’s morphological complexity and the effect of syntactic complexity for predicates, supporting the competition model. Predictor coefficients revealed broad consistency across all languages, along with variability that reflected language family classifications.

Keywords: language acquisition; word learning; vocabulary; cross-linguistic; age of acquisition

Introduction

Why are some words learnt earlier than others? Despite the variation in the structure of different languages, children demonstrate remarkably similar early lexical development across languages, with highly overlapping first words (Frank, Braginsky, Yurovsky, & Marchman, 2021; Tardif et al., 2008), and similar emergence of lexical categories (Caselli et al., 1995). The order of lexical development can thus be used as a tool to understand the commonalities and differences in the factors that predict word learning across languages and lexical categories.

Such questions have led to a productive line of inquiry, with previous research demonstrating that, in English, words learnt earlier tend to be more frequent (Goodman, Dale, & Li, 2008), be more iconic (Perry, Perlman, & Lupyan, 2015), and appear in shorter utterances (Swingley & Humphrey, 2018). The conclusions of these studies, however, were limited in generalisability as they considered different sets of predictors, and mostly focused only on English-learning children. To have a more comprehensive understanding of early language learning, it is important to capture two dimensions of variability: across languages, and across levels of linguistic representation (e.g., phonological, morphological, syntactic, and

semantic). These dimensions jointly provide information about how different sources of information may be differentially relevant across languages of different structures.

One study that has adopted this approach comes from Braginsky, Yurovsky, Marchman, & Frank (2019), who conducted a larger scale cross-linguistic study across 10 languages, using a larger set of predictors to determine their independent contributions to word learning (see also Frank et al., 2021). Overall, they found strong consistent effects for predictors such as frequency and concreteness, but little to no effect of predictors such as valence. Nonetheless, there are two crucial limitations to their set of analyses. First, among the predictors they considered, most were distributional (unigram frequency, frequency as sole or last word in utterance) or semantic (concreteness, babiness, valence, arousal), with only one phonological predictor (length in phonemes) and one syntactic predictor (mean length of utterance). Although there are cross-linguistic and cultural differences in word distributions in naturalistic speech, frequencies are remarkably consistent, especially for fundamental vocabulary (Calude & Pagel, 2011), and semantic representations also tend to exhibit similar organisations cross-linguistically, at least within semantic domains (Lewis, Cahill, Madnani, & Evans, 2023). It is thus important to consider the full range of levels of linguistic representation, particularly the levels which are more likely to exhibit larger variation across languages, namely morphological and syntactic factors.

Second, of the 10 languages examined by Braginsky et al. (2019), all but one (Turkish) were Indo-European. As such, it is possible that some of the consistency that they observed was in fact due to structural similarities among Indo-European languages, as opposed to underlying patterns in language learning across languages regardless of language family. To have a more generalisable understanding of the factors driving early word learning, it is imperative to study a more diverse set of languages, representing more language families (see Kidd & Garcia, 2022).

Hence, the present study aims to capture a more comprehensive view of early word learning by including a broad set of predictors encompassing phonological, morphological, syntactic, and semantic levels of representa-